2025/06/08

Vu H. N. Phan

Curriculum Vitae

Portland, Oregon
vuphan314.github.io
vuphan314
vuphan314

Vu Phan is an Intellectual Property validation engineer at Intel Corporation; he performs pre-Silicon Formal Verification of computer-hardware IP. Vu earned a Ph.D. degree in Computer Science at Rice University; his advisor was Prof. Moshe Vardi.

Work Experience

2022/10 –now **IP Validation Engineer**, *Intel Corporation*, Full-time, Portland, Oregon (hybrid).
- Perform Formal Property Verification of System-on-a-Chip IP
- Find bugs by checking Register-Transfer Level implementations against architectural specifications

Skills: Computer Architecture, SystemVerilog, Tool Command Language

2017/08 –2022/08 **Graduate Research Assistant**, *Rice University*, Full-time, Houston, Texas (on-site).
- Designed and implemented algorithms to solve stochastic satisfiability with Prof. Moshe Vardi

Skills: C++, Python

2021/05 –2021/08 **IP Validation Engineer**, *Intel Corporation*, Intern, Dallas, Texas (remote).
- Developed software to benchmark Formal Property Verification platforms

Formal Education

2020/01 –2022/08 **Doctor of Philosophy in Computer Science**, *Rice University*, GPA 3.63/4.00.
- Published papers [DPV20a; DPV21] thesis [Pha22]

2017/08 –2019/12 **Master of Science in Computer Science**, *Rice University*, GPA 3.60/4.00.
- Published paper [DPV20b] and thesis [Pha19]

2014/08 –2017/07 **Bachelor of Science in Computer Science & Math**, *Texas Tech University*, GPA 4.00/4.00.
- Published paper [Pha18]

Software Projects

2019 –2021 **DPMC**, *Dynamic-Programming Model Counter*, `https://github.com/vardigroup/DPMC`.
- Implemented a weighted-#SAT framework with Jeffrey Dudek

2018 –2019 **ADDMC**, *Algebraic-Decision-Diagram Model Counter*, `https://github.com/vardigroup/ADDMC`.
- Implemented a weighted-#SAT solver

2016 –2018 **LED**, *Language of Effective Definitions*, `https://vuphan314.github.io/LED`.
- Translated the literate-programming language `LED` into `SequenceL` and `LaTeX`

2015 –2016 **L**, *Logic*, `https://github.com/iensen/LtoASPtranslator`.
- Translated the logic-programming language `L` into `Answer Set Prolog` with Evgenii Balai
- Innovated CertWare Safety Case Workbench Software with five others (NASA NTR: LAR-18868-1)

Computing Courses

2017 –2021 **Graduate Level**.

1. Reasoning about Software
2. Bioinformatics: Sequence Analysis
3. Statistical Machine Learning
4. Computer Systems Architecture
5. Multi-Core Computing
6. Compiler Construction
7. Artificial Intelligence
8. Programming Languages
9. Automated Program Verification
10. Logic in Computer Science

2015 –2017 **Undergraduate Level**.

1. Operating Systems
2. Database Systems
3. Computer Architecture
4. Programming Languages
5. Software Engineering
6. Algorithms
7. Object-Oriented Programming
8. Computer Organization and Assembly Language
9. Modern Digital System Design
10. Data Structures
11. Automata
12. Programming Principles

Bibliography

Vu Phan is the correspondence author of the following publications (authors are sorted by surnames).

Theses

[Pha22] Vu H. N. Phan. "Quantitative Reasoning on Hybrid Formulas with Dynamic Programming". PhD thesis. Rice University, 2022. URL: https://repository.rice.edu/items/2e464125-244d-431b-b998-612f0dc2b41a.

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: https://repository.rice.edu/items/a1a5e73d-a001-44ca-9730-25a7277c8af1.

Conference Papers

[DPV21] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ProCount: Weighted Projected Model Counting with Graded Project-Join Trees". In: *Conference on Theory and Applications of Satisfiability Testing (SAT)*. 2021. URL: https://kasekopf.github.io/papers/sat21_procount.pdf.

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: https://arxiv.org/abs/2008.08748.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: https://arxiv.org/abs/1907.05000.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: https://arxiv.org/abs/1809.09319.